Topological Bypass of the Ford–Roman Bound: K-Theoretic Protection of Vacuum Energy Sign and a Testable Prediction for DN Casimir Scaling

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Three experimentally established facts form a logical chain: (i) the Casimir force obeys ζ-regularized vacuum energy ($\zeta(-1) = -1/12$ in 1D, $\zeta(-3) = 1/120$ in 3D); (ii) changing boundary conditions from Dirichlet–Dirichlet (DD) to Dirichlet–Neumann (DN) reverses the Casimir energy sign, corresponding to $\zeta \to \zeta_{\neg 2}$; (iii) the observed dark energy density is consistent with $\rho_\Lambda \propto \zeta(-3)$. The natural fourth step—that the vacuum energy density itself can be sign-reversed by $\zeta \to \zeta_{\neg 2}$—is a straightforward extrapolation that no established physics contradicts.

We argue that the localization $\mathbb{Z} \to \mathbb{Z}[1/2]$ underlying the DN configuration introduces a topological winding number ($K_1(\mathbb{Z}[1/2]) = \mathbb{Z}/2 \times \mathbb{Z}$) that *protects* the reversed sign. If this protection is physical, it provides a mechanism to bypass the Ford–Roman quantum energy inequalities, which assume continuous deformations. A topological (discrete) phase transition is outside the scope of these inequalities, just as BCS superconductivity is outside the scope of Drude resistivity.

We predict a single observable consequence: the DN Casimir energy density deviates from $1/a^4$ decay at a critical separation a_c, plateauing at a constant value. This can be tested with existing Si-MEMS apparatus (Princeton/Nature Photonics 2017).

THREE ESTABLISHED FACTS

We begin with three experimentally or observationally confirmed results, each involving zeta-function regularization.

Fact 1. The Casimir force between parallel conducting plates (DD boundary conditions) is

$$F/A = -\frac{\pi^2 \hbar c}{240\, a^4}, \tag{1}$$

where $240 = 2/\zeta(-3)$. Measured to $< 1\%$ accuracy [1, 2]. This confirms that $\zeta(-3) = 1/120$ is a physically realized quantity, not merely a mathematical convenience.

Fact 2. Changing to DN boundary conditions (one perfectly conducting, one perfectly magnetically conducting wall) reverses the sign: $E_{DN} \propto \zeta_{\neg 2}(-1) = +1/12 > 0$ (repulsive) [3, 4]. Geometric realizations of Casimir repulsion have been demonstrated [5, 6], confirming that the sign of vacuum energy is controllable.

Fact 3. The observed dark energy density $\rho_\Lambda \approx 6 \times 10^{-10}$ J/m^3 is consistent with $\rho_\Lambda \propto \rho_P\, \zeta(-3)\, (\ell_P/R_H)^2$, where ρ_P is the Planck density and R_H the Hubble radius [7].

The Fourth Step. If $\zeta(-3)$ governs the vacuum energy (Facts 1,3) and its sign can be reversed by $\zeta \to \zeta_{\neg 2}$ (Fact 2), then the vacuum energy density itself should be reversible: $\rho_\Lambda \to \rho_\Lambda \times \zeta_{\neg 2}(-3)/\zeta(-3) = -7\rho_\Lambda < 0$. No established physics forbids this extrapolation.

WHY THIS IS NOT TRIVIALLY ACHIEVED

The sign reversal in Fact 2 is a *boundary effect*: the Casimir energy density $\rho \propto \pm 1/a^4$ depends on the plate separation a and vanishes as $a \to \infty$. For applications to dark energy or spacetime engineering, one needs a *bulk effect*: a constant ρ independent of a.

Standard QED predicts $\rho \propto 1/a^4$ for *all* boundary conditions. The sign changes but the scaling does not. At macroscopic separations, the effect is negligible ($\rho \sim 10^{-24}$ J/m^3 at $a = 1$ m).

The Ford–Roman quantum energy inequalities [8, 9] further constrain negative energy: $|\Delta E| \cdot \Delta t \lesssim \hbar$, preventing sustained macroscopic negative energy in any state reachable by continuous deformation from the vacuum.

THE TOPOLOGICAL BYPASS

We propose that these constraints can be circumvented by a *topological phase transition* in the vacuum.

The DD→DN transition corresponds to the algebraic localization $\mathbb{Z} \to \mathbb{Z}[1/2]$, which changes the K-theory:

$$K_1(\mathbb{Z}) = \mathbb{Z}/2 \xrightarrow{\text{localize}} K_1(\mathbb{Z}[1/2]) = \mathbb{Z}/2 \times \mathbb{Z}. \tag{2}$$

The emergent $\mathbb{Z}$ is a winding number. In condensed-matter physics, analogous winding numbers (SSH model [10], Chern numbers in quantum Hall systems) protect macroscopic observables:

- The quantized Hall conductance $\sigma_{xy} = \nu e^2/h$ is *independent of sample size*, even though it arises from edge states (a boundary effect). The bulk–edge cor-

respondence promotes a boundary observable to a bulk-protected quantity.

- The superconducting gap in BCS theory is *independent of the coherence length* once the condensate forms. The gap is topologically protected (by the U(1) phase winding), not set by microscopic scales.

We conjecture an analogous mechanism for the vacuum: the winding number in (2) protects the *sign* of the vacuum energy, and once the topological transition occurs, the energy density becomes a *bulk* quantity (constant, not $1/a^4$).

Why Ford–Roman does not apply. The quantum energy inequalities are derived for states reachable by *continuous* deformation from the Minkowski vacuum. A topological transition (K_1 winding number jumping from 0 to 1) is *discrete* and outside this class. Just as BCS superconductivity violates the Drude prediction of finite resistivity (by a phase transition, not by continuous improvement of conductivity), the topologically protected vacuum could violate the Ford–Roman bound by a phase transition, not by continuous accumulation of negative energy.

THE PREDICTION

If the vacuum energy sign is topologically protected, the DN Casimir energy density must deviate from $1/a^4$ at a critical separation a_c:

	Standard QED	Topological
$a < a_c$	$\rho \propto 1/a^4$	$\rho \propto 1/a^4$
$a > a_c$	$\rho \propto 1/a^4$	$\rho \to$ const
Force exponent α	5	$5 \to 0$ at a_c

The prediction is the *existence* of the deviation. The value of a_c is not predicted; it could range from nanometers to macroscopic scales.

Experimental test. Measure the DN Casimir force $F(a)$ over two decades in a and fit $F \propto a^{-\alpha}$. Any confirmed $\alpha \neq 5$ in the DN configuration, absent in DD, constitutes evidence for topological protection. The Princeton Si-MEMS apparatus [6], which has already demonstrated geometric Casimir repulsion at ~ 100 nm, is a suitable platform.

IMPLICATIONS

If confirmed, the topological protection of vacuum energy sign has profound consequences:

1. *Controlled dark energy.* The vacuum energy density becomes a switchable quantity, controlled by boundary topology (DD vs. DN).

2. *Macroscopic negative energy.* A topologically protected constant $\rho < 0$ filling a volume V gives $E = -|\rho|V$, scaling with volume rather than surface area.

3. *Bypass of Ford–Roman.* Sustained negative energy becomes possible via a topological phase, not a continuous deformation.

4. *Spacetime engineering.* Negative energy of sufficient magnitude enables exotic spacetime geometries (warp metrics [11], traversable wormholes [12]).

We emphasize that these implications are contingent on the detection of the scaling deviation. The paper's contribution is to identify a *single, falsifiable prediction*—$\alpha \neq 5$ for DN Casimir—that distinguishes the topological scenario from standard QED.

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